NAND to Tetris-I

Lab 1: Digital Logic & ALU Design

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Q1. Q-5 (10 marks)

CHIP ALU { IN X[16],Y[16] ZX, NX, ZY, NY, F, NO, OUT out[16], zr, ng, PARTS: MUX16(a=x,b=false,sel=zx,out=x1): not16(in=x1,out=x1): MUX16(a=x1,b=nx1,sel=nx,out=x2): MUX16(a=y,b=false,sel=zy,out=y1): not16(in=y1.out=y1): MUX16(a=fAnd,b=fAnd,sel=f,out=fout): not16(in=fout,=nfout): MUX16(a=fout,b=nfout,sel=no,out=out,out(0,7)=!out(8,15)=high8,out[15]=ng): or8way(in=low8,out=orLow): or8way(in=high8,out=orhigh): or(a=orLow,b=orhigh,out=notzero): Not(in=notZero,out=zr); }